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# On the Irreducibility Conjecture Based on Computer Calculation for Yablonskii-Vorobiev Polynomials Which Give a Rational Solution of the Toda Equation of Painlevé-II Type*

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An interesting sequence of polynomials which were first introduced by A. I. Yablonskii and A. P. Vorobiev to describe rational solutions of the Painlevé-II equation is considered. It also gives a special rational solution of the Toda equation. All our polynomials have integral coefficients. By computer calculation we proved rigorously that the first 24 of our polynomials are irreducible. It is natural to suppose that all our polynomials are irreducible. But our numerical results show that it is a very difficult mathematical problem to determine whether or not all our polynomials are irreducible or not. E. R. Berlekamp's algorithm was used for numerical calculation instead of usual Kronecker's algorithm.

*Key words*: Toda equation, Painlevé-II equation, rational solution, irreducible polynomial

## 1. Introduction

Recently many authors discovered a misterious link among the *KdV* equation, the modified *KdV* equation, the Toda equation and the Painlevé-II equation. All the above equations are very important and fundamental in soliton theory. Some special rational solutions of the equations are given by a sequence of special polynomials, $P_n(t)$ $(n=0, 1, 2, \cdots)$, which are given by (1.1) and (1.2). We call them Yablonskii-Vorobiev polynomials because A. I. Yablonskii [1] and A. P. Vorobiev [2] first introduced this sequence of polynomials to describe the rational solutions of the Painlevé-II equations. At that time they did not mention the Toda equation. It was discovered by Toda in 1966 [3]. But, to our great surprise, the Toda equation with two time variables can be seen in the famous book of G. Darboux [4].

Our polynomials, $P_n(t)$, all have integral coefficients. The problem of irreducibility of $P_n(t)$ is treated especially by the aid of computer calculation. Our results show that E. R. Berlekamp's algorithm [5] is very efficient when we wish to determine whether or not a given polynomial with integral coefficients is irreducible. But when we wish to determine whether or not all $P_n(t)$ are irreducible our results (computer calculation) show that it is a very difficult mathematical problem. In general it is very hard to show theoretically that all members of a certain sequence of infinitely many

* This paper is dedicated to late Professor T. Miyata.

polynomials are irreducible. In this direction we know only a few successful examples. The Laguerre polynomials and the Hermite polynomials are all irreducible. The Tchebycheff polynomial, $T_n(x)$, is irreducible if $n$ is odd and prime or if $n=2^m$ ($m$ is a nonnegative integer). It is also well known that the cyclotomic polynomials are all irreducible.

Another interesting problem of distribution of zeros of $P_n(t)$ is considered in [6]. The maximal modulus of zeros of $P_n(t)$ is $O(n^{2/3})$ as $n$ tends to infinity.

Following Yablonskii and Vorobiev let us introduce an interesting sequence of polynomials:

$$P_0=P_1=1\,,\qquad P_2=t\,,\qquad P_3=t^3+4\,,\qquad P_4=t^6+20t^3-80\,,$$

$$P_5=t^{10}+60t^7+11200t\,,$$

$$P_6=t^{15}+140t^{12}+2800t^9+78400t^6-3136000t^3-6272000\,,$$

$$P_7=t^{21}+280t^{18}+18480t^{15}+627200t^{12}-17248000t^9$$
$$+1448832000t^6+19317760000t^3-38635520000\,,$$

$$P_8=t^{28}+504t^{25}+75600t^{22}+5174400t^{19}+62092800t^{16}$$
$$+13039488000t^{13}-828731904000t^{10}-49723914240000t^7$$
$$-3093932441600000t,\cdots$$

by the recurrence relation,

$$P_0=P_1=1\,, \tag{1.1}$$

$$P_{n-1}P_{n+1}=tP_n^2+4P_n'^2-4P_nP_n''\,,\quad \left(\,'=\frac{d}{dt}\right) \tag{1.2}$$

$$P_{-n}=P_{n+1}\,,\qquad n=1,2,3,\cdots \tag{1.3}$$

to describe the rational solutions,

$$q_n=(\log P_n/P_{n+1})' \tag{1.4}$$

of the Painlevé-II equations,

$$q_n''=2q_n^3+tq_n+n\,. \tag{1.5}$$

If we define $p_n$ by

$$p_n=-P_nP_{n+2}/4P_{n+1}^2=(\log P_{n+1})''-t/4 \tag{1.6}$$

then $\{q_n, p_n\}$ satisfies the Toda equation,

$$q_n'=p_{n-1}-p_n\,,\qquad p_n'=p_n(q_n-q_{n+1})\,. \tag{1.7}$$

$P_n$ has the form of

$$P_n = \sum_{j=0}^{f(n)} P_{n,j} t^{d(n)-3j} \tag{1.8}$$

where $d(n) = n(n-1)/2$ and $f(n) = [n(n-1)/6]$[1]. $P_n$ is a polynomial of degree $d(n)$ with integral coefficients ($P_{n,0} = 1$, $P_{n,f(n)} \neq 0$). If we introduce associated polynomials $F_n(x, t)$ by

$$F_n(x, t) = \sum_{j=0}^{f(n)} P_{n,j} (3t)^j x^{3(f(n)-j)} \tag{1.9}$$

then,

$$u_n(x, t) = g(n+1)x^{-2} - \left(\frac{\partial}{\partial x}\right)^2 \log F_{n+1}(x, t) \tag{1.10}$$

and

$$v_n(x, t) = (g(n) - g(n+1))x^{-1} + \frac{\partial}{\partial x} \log (F_n(x, t)/F_{n+1}(x, t)) \tag{1.11}$$

($g(n) = 1$ if $n \equiv 2 \pmod 3$, $= 0$ otherwise) are the rational similarity solutions of the *KdV* equation,

$$u_t - 12uu_x + u_{xxx} = 0 \tag{1.12}$$

and the modified *KdV* equation,

$$v_t - 6v^2 v_x + v_{xxx} = 0 \tag{1.13}$$

respectively [7], [8].

By computer calculation we proved that $P_n$ ($n \leqslant 23$) are essentially irreducible. That is, the corresponding $F_n(x)$ (given by (2.1)) are irreducible. So it is probable that they are all essentially irreducible. If we can show that all $P_n$ are irreducible it means that the representation (1.4) is the simplest. That is, each of the rational solutions of the Painlevé-II equations can be expressed by two irreducible polynomials.

## 2. Recurrence Relations for the Coefficients

Instead of $P_n$ we treat $F_n$ which are defined by

$$P_n(t) = 4^{f(n)} t^{g(n)} F_n(t^3/4) \tag{2.1}$$

where $f(n)$ and $g(n)$ are given in section 1, but all $F_n$ are different from those in section 1. All $F_n$ have the form of

$$F_n(x) = \sum_{j=0}^{f(n)} F_{n,j} x^{f(n)-j} \tag{2.2}$$

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[1] For any real number $x$ $[x]$ represents its integral part.

where

$$F_{n,j}=4^{-j}P_{n,j}\,,\qquad 0\leqslant j\leqslant f(n)\,. \tag{2.3}$$

It follows from (1.1) and (1.2) that the coefficients $F_{n,j}$ satisfy the following recurrence relations:

$$F_{n,0}=1\,,\qquad F_{n,1}=(n-2)(n-1)n(n+1)/24\,,\qquad n=0,\,1,\,2,\,\cdots \tag{2.4}$$

$$\begin{aligned}F_{n+1,j}=&-F_{n-1,j}-\sum_{k=1}^{j-1}F_{n-1,j-k}F_{n+1,k}+2A_{n,j}\\&+\operatorname{mod}(j+1,2)F_{n,[j/2]}^{2}+\{n(n-1)-3(j-1)\}A_{n,j-1}\\&+\{n(n-1)/2-3[(j-1)/2]\}\operatorname{mod}(j,2)F_{n,[(j-1)/2]}^{2}-9B_{n,j-1}\,,\\&\qquad\qquad j=2,\,3,\,4,\,\cdots,\quad n=1,\,2,\,3,\,\cdots,\end{aligned} \tag{2.5}$$

$$A_{n,0}=0\,,\qquad A_{n,1}=F_{n,1}\,,\qquad A_{n,2}=F_{n,2}\,, \tag{2.6}$$

$$B_{n,0}=0\,,\qquad B_{n,1}=F_{n,1}\,,\qquad B_{n,2}=4F_{n,2}\,, \tag{2.7}$$

$$A_{n,j}=F_{n,j}+\sum_{k=1}^{[(j-1)/2]}F_{n,j-k}F_{n,k}\,, \tag{2.8}$$

$$B_{n,j}=j^{2}F_{n,j}+\sum_{k=1}^{[(j-1)/2]}(j-2k)^{2}F_{n,j-k}F_{n,k}\,,\qquad j=3,\,4,\,5,\,\cdots,\ n=0,\,1,\,2,\,\cdots. \tag{2.9}$$

$$F_{n,j}=0\qquad\text{if}\quad j>f(n)\,,\quad n=0,\,1,\,2,\,\cdots, \tag{2.10}$$

where $\operatorname{mod}(j,2)=0$ if $j$ is even, $=1$ if $j$ is odd. From these relations it follows that all $F_{n,j}$ are integers.

## 3. Irreducibility of $F_0\sim F_9$

$F_0=F_1=F_2=1$, $F_3=x+1$, $F_4=x^2+5x-5$ are trivially irreducible. $F_5=x^3+15x^2+175$ is irreducible mod (2) so it is irreducible globally. Another reasoning is also possible. We can apply G. Dumas' method [9] to $F_5(x+5)=x^3+2\cdot3\cdot5x^2+3^2\cdot5^2x+3^3\cdot5^2$. Drawing Newton's polygon we immediately see that $F_5(x+5)$ is irreducible. So $F_5(x)$ is also irreducible. We can also apply Dumas' method to

$$F_6=x^5+5\cdot7x^4+5^2\cdot7x^3+5^2\cdot7^2x^2-2\cdot5^3\cdot7^2x-5^3\cdot7^2\,,$$

$$\begin{aligned}F_7=&x^7+2\cdot5\cdot7x^6+3\cdot5\cdot7\cdot11x^5+2^3\cdot5^2\cdot7^2x^4-5^3\cdot7^2\cdot11x^3\\&+3\cdot5^3\cdot7^3\cdot11x^2+2\cdot5^4\cdot7^3\cdot11x-5^4\cdot7^3\cdot11\end{aligned}$$

and

$$\begin{aligned}F_9=&x^{12}+2\cdot3\cdot5\cdot7x^{11}+3\cdot5\cdot7\cdot11\cdot13x^{10}+2\cdot5^2\cdot7^2\cdot11\cdot19x^9\\&+3^2\cdot5^3\cdot7^2\cdot11\cdot13x^8+3^2\cdot5^2\cdot7^3\cdot11^2\cdot13x^7-2^3\cdot3\cdot5^3\cdot7^3\cdot11^2\cdot13x^6\end{aligned}$$

$$+3^2\cdot 5^3\cdot 7^3\cdot 11^2\cdot 13\cdot 41x^5+3^2\cdot 5^4\cdot 7^5\cdot 11^2\cdot 13^2x^4+2^2\cdot 5^5\cdot 7^4\cdot 11^3\cdot 13^2x^3$$
$$+3\cdot 5^5\cdot 7^5\cdot 11^3\cdot 13^2x^2-3\cdot 5^6\cdot 7^5\cdot 11^3\cdot 13^2x-5^6\cdot 7^5\cdot 11^3\cdot 13^2 .$$

We can conclude that they are all irreducible.

## 4. Computational Results

It becomes more difficult to conclude that $F_n$ is irreducible as $n$ increases because $F_{n,j}$ become too large for large $n$ and $j$. In our previous work [8] we gave a theoretical bound $|F_{n,j}|\leqslant 4^{-j}(7n)^{4j}$ for these coefficients. When we have a large computer we fortunately have the very good algorithm of E. R. Berlekamp [5]. Using his algorithm we can enumerate the number of irreducible factors modulo a prime number efficiently.

Numerical calculation was done by a computer at the Computer Center of Kyushu University. We arrived at the following conclusion:

THEOREM. *The number of irreducible factors of $F_n$ is one modulo the following primes $p_n$:*

$$p_5=p_6=2,\quad p_7=19,\quad p_8=17,\quad p_9=53,\quad p_{10}=167,$$
$$p_{11}=251,\quad p_{12}=2,\quad p_{13}=41,\quad p_{14}=37,\quad p_{15}=211,$$
$$p_{16}=223,\quad p_{17}=283,\quad p_{18}=191,\quad p_{19}=239,\quad p_{20}=1447,$$
$$p_{21}=149,\quad p_{22}=773,\quad p_{23}=211 .$$

*$F_n$ ($n\leqslant 23$) are irreducible.*

$p_{20}\sim p_{23}$ were obtained by the help of Professor Yoshiaki Hashimoto of Nagoya City University.

## 5. Appendix

The coefficients of $P_9$ and $P_{10}$ were obtained by Professor Matsutaro Noda of Ehime University by microcomputer.

$P_{9,1}=840$ $\quad$ $P_{9,7}=407736096768000$

$P_{9,2}=240240$ $\quad$ $P_{9,8}=126696533483520000$

$P_{9,3}=32771200$ $\quad$ $P_{9,9}=1769729356595200000$

$P_{9,4}=2018016000$ $\quad$ $P_{9,10}=37164316488499200000$

$P_{9,5}=124309785600$ $\quad$ $P_{9,11}=-743286329769984000000$

$P_{9,6}=-6629855232000$ $\quad$ $P_{9,12}=-991048439693312000000$

$$P_{10,1} = 1320$$

$$P_{10,2} = 646800$$

$$P_{10,3} = 160160000$$

$$P_{10,4} = 21525504000$$

$$P_{10,5} = 1876485811200$$

$$P_{10,6} = 46132742656000$$

$$P_{10,7} = 8403341506560000$$

$$P_{10,8} = 483515342069760000$$

$$P_{10,9} = -31810885 1847987200000$$

$$P_{10,10} = -184303154654537 31840000$$

$$P_{10,11} = -558505348189165977600000$$

$$P_{10,12} = 963695502757776588 8000000$$

$$P_{10,13} = -404752111158266167296000000$$

$$P_{10,14} = -4047521111582661672960000000$$

$$P_{10,15} = 539669481544354889728000000 0$$

Calculation of the coefficients of $P_n$ for sufficiently small $n$ by a large computer is quite easy.

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